INCH-POUND

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MILITARY SPECIFICATION

TRAINER, PLASTIC FLUID CONTAINER ASSEMBLY

This specification is approved for use l the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of plastic fluid container assembly trainer for the M13 Decontaminating Apparatus, Portable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6910

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STANDARDS

MILITARY

MIL-STD-1168 - Ammunition Lot Numbering.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-51-598 - Plastic Fluid Container Assembly, Trainer

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-51-598 - Plastic Fluid Container Assembly, Trainer

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229-7210.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-51-598 and on the subsidiary drawings shall conform to the specifications and drawings listed thereon.

3.1.2 Components. All components of the trainer shall conform to the specifications and drawings listed on Drawing 5-51-598 and subsidiary drawings.

3.2 Manufacture and assembly. The trainer shall be manufactured and assembled in accordance with Drawing 5-51-598.

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3.3 **Performance.**

3.3.1 **Leakage.** The trainer shall not leak when subjected to a minimum internal pressure of 3 pounds per square inch gage (psig), for a minimum of 15 seconds when tested in accordance with 4.4.6.1.

3.3.2 **Drop strength.** The trainer shall not leak when dropped from a height of 4 feet onto solid concrete so as to strike on a bottom corner. The container shall be filled to 98 percent capacity with water, when tested in accordance with 4.4.6.2.

3.3.3 **Training container (Drawing 5-51-599) lifting handle.** The lifting handle shall support a load of at least 150 pounds for a minimum of 5 seconds without distortion of the handle when tested in accordance with 4.4.6.3.

3.4 **First article.** Unless otherwise specified in the contract, a first article sample of trainers shall be produced in accordance with this specification, for examination and test (see 4.3 and 6.2), prior to the start of regular production and each time a new or reworked mold is used during regular production.

3.5 **Workmanship.** The trainer shall be free from cracks, dents, sharp edges, and foreign matter such as dirt, grease, and oily or viscous materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.4.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in 4.3.3 or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.3.2 Sample. The first article sample shall consist of 100 trainers manufactured and packaged using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.3 Inspection procedures.

4.3.3.1 For examination. Twenty trainers from the first article sample shall be taken at random and examined for all characteristics in the classification of characteristics, except leakage, drop strength and lifting handle strength (see 4.3.3.2), using the inspection

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methods specified therein. The sampling and acceptance criteria of the classification of characteristics do not apply.

4.3.3.2 For test. Trainers, taken at random from the first article sample, shall be subjected to the tests specified in table I.

TABLE I. First article tests

First article test	Number of first article samples
Leakage (4.4.6.1)	20
Drop strength (4.4.6.2)	5
Lifting handle strength (4.4.6.3)	20*

*Inspect training containers (Drawing 5-51-599) prior to assembly.

4.3.3.3 For packaging. Twenty packaged trainers from the first article sample shall be taken at random and examined in accordance with the classification of characteristics for packaging. The sampling and acceptance criteria of the classification of characteristics do not apply.

4.3.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of trainers (see 6.2) shall consist of those like items produced at the same place utilizing the same batches of materials, lot of components, process runs, fabrication techniques, assembly methods, tools, equipment and facilities, but shall not exceed one month's production. No more than one lot or batch (see 6.3) of material shall be used in any one lot of trainers and a new lot interfix number shall be established when a new or reworked mold is used. Each lot shall be controlled and identified in accordance with MIL-STD-1168.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and, when specified, table II. Samples shall be selected at random.

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TABLE II. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.4.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the

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item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table II for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(a)	Plastic Fluid Container Assembly, Trainer		5-51-598
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Leakage	Table II, Level IV	3.3.1	4.4.6.1
102	Torque correct	Table II, Level VII	3.1.2	CE
103	Drop strength	Table II, Level XI	3.3.2	4.4.6.2*
Minor				
201	Specified components correctly assembled	Table II, Level IX	3.1.2	VI
202	Labels evident, correct and correctly located	Table II, Level IX	3.1.2	VI
203	Crimping correct	Table II, Level IX	3.1.2	CE
204	Workmanship	Table II, Level IX	3.5	VI
205	Protective finish evident and correct	Table II, Level IX	3.1.1	VI

NOTES:
* Destructive test
CE – Commercial inspection equipment
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 5-51-599
4.4.5(b)	Container, Training, Plastic		NEXT HIGHER ASSY 5-51-598

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Lifting handle strength	Table II, Level VI	3.3.3	4.4.6.3
102	Height (18.10)	Table II, Level VII	3.1.2	CE
103	Length (9.99) and Width (6.57)	Table II, Level VII	3.1.2	CE
104	Radius (2.36) (2 places)	Table II, Level VII	3.1.2	CE
105	Torque correct	Table II, Level VII	3.1.2	CE
106	Wall thickness	Table II, Level VII	3.1.2	CE
107	Height (18.45 maximum)	Table II, Level VII	3.1.2	CE
108	Dimension (1.00)	Table II, Level VII	3.1.2	CE
Minor				
201	Specified components correctly assembled	Table II, Level IX	3.1.2	VI
202	Workmanship	Table II, Level IX	3.5	VI

NOTES:
CE – Commercial inspection equipment
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(c)	Packaging		SPI P5-51-598
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Filler pads evident and correct	Table II, level IX	5.1	VI
102	Marking evident and correct	Table II, level IX	5.1	VI
103	Banding straps evident, unbroken, tight and correctly located	Table II, level IX	5.1	VI
104	Container size correct	Table II, level IX	5.1	VI
Minor				
201	Pallet correct and undamaged	Table II, level XI	5.1	VI

NOTES:
CE – Commercial inspection equipment
VI – Visual inspection

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4.4.6 Tests. Unless otherwise specified, tests shall be performed at room temperature, 20°-30°C (68°-86°F) as follows:

4.4.6.1 Leakage. With the container openings sealed and with the specified air pressure introduced into the container through a fitting, submerge the container in water containing a wetting agent to a depth of no more than 3 inches below the surface for the specified time and observe for leakage. A steady stream of bubbles shall be an indication of leakage.

4.4.6.2 Drop strength. Fill the container as specified and seal the openings. Drop the container as specified on a bottom corner.

4.4.6.3 Lifting handle. Secure the container and apply the specified static load uniformly over the handle for the specified time. Examine lifting handle and container.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P5-51-598.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The plastic fluid container assembly trainer is intended to be used as a training container for the M13 Portable, 14 Liter, Decontaminating Apparatus. The training container is to be filled with water or other approved simulant for DS2.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(c) Level of preservation, packaging, packing, palletization required and shelf life code (see section 5).
(d) First article:

(1) First Article Inspection and Report requirements.
(2) Delivery schedule for First Article Report.
(3) Destination and delivery schedule of First Article Samples when testing is performed by the Government.
(4) First Article Test report approval time.

(e) Production:

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(1) Destination and delivery schedule requirements.
(2) Production and inspection report requirements.

6.3 Batch. A lot or batch is defined as that quantity of material which has been manufactured at essentially the same time by some unit chemical process or which has been subjected to physical mixing of the same ingredients by the same unit process to make the final product homogeneous and uniform and is defined as a lot or batch by the manufacturer.

6.4 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing.

Chemical agents
Decontamination
Decontamination equipment
Training container

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6910-A097

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